

BRIDGE: Benchmarking Large Language Models for Understanding Real-world Clinical Practice Text

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Abstract

Large language models (LLMs) hold great promise for medical applications and are evolving rapidly, with new models being released at an accelerated pace. However, benchmarking on large-scale real-world data such as electronic health records (EHRs) is critical, as clinical decisions are directly informed by these sources, yet current evaluations remain limited. Most existing benchmarks rely on medical exam-style questions or PubMed-derived text, failing to capture the complexity of real-world clinical data. Others focus narrowly on specific application scenarios, limiting their generalizability across broader clinical use. To address this gap, we present BRIDGE, a comprehensive multilingual benchmark comprising 87 tasks sourced from real-world clinical data sources across nine languages. It covers eight major task types spanning the entire continuum of patient care across six clinical stages and 20 representative applications, including triage and referral, consultation, information extraction, diagnosis, prognosis, and billing coding, and involves 14 clinical specialties. We systematically evaluated 95 LLMs (including DeepSeek-R1, GPT-4o, Gemini series, and Qwen3 series) under various inference strategies. Our results reveal substantial performance variation across model sizes, languages, natural language processing tasks, and clinical specialties. Notably, we demonstrate that open-source LLMs can achieve performance comparable to proprietary models, while medically fine-tuned LLMs based on older architectures often underperform versus updated general-purpose models. The BRIDGE and its corresponding leaderboard serve as a foundational resource and a unique reference for the development and evaluation of new LLMs in real-world clinical text understanding.

Keywords: Large Language Models, Electronic Healthcare Records, Multilingual, Real-world clinical data, Benchmark.

Introduction

Recent advances in large language models (LLMs) have demonstrated a transformative potential in improving healthcare delivery and clinical research.¹ By combining extensive pretraining on vast corpora with supervised instruction tuning across diverse tasks,^{2,3} LLMs exhibit exceptional capabilities in textual understanding, generation, and reasoning,⁴⁻⁶ and show promise in medical

applications. The prompt-based instruction provides an intuitive and easy-to-use approach to interact with LLMs on diverse tasks. For clinicians, LLMs can support the drafting of clinical documentation^{7,8} and assist in clinical decision-making⁹, enhancing efficiency and reducing workload burdens.¹⁰ LLMs also offer the promise to benefit patients by providing simple interpretations of complex medical information¹¹ and personalized preventive advice,¹² promoting patient engagement, treatment adherence, and overall disease management.^{13,14} Consequently, LLMs hold significant promise for improving the quality, cost and accessibility of healthcare services worldwide.¹⁵ However, concerns persist regarding the reliability and clinical validity of LLM-generated outputs,¹⁶ particularly given the high diversity of clinical tasks, specialties, and languages.¹⁷ Moreover, LLMs are rapidly evolving, with diverse models emerging weekly, including proprietary and open-source, medical specific and general-purpose, and varying in size. The general-purpose nature of LLM usage complicates institutional model selection and comparison. While clinical trials are necessary for the most high-risk cases, they are slow, expensive, and cannot possibly investigate every single use case. Clinical benchmarks – automated, timely, and systematic evaluations of model performance – remain essential for clinicians, patients, health systems, and regulators, providing both understanding of the usability and trustworthiness of LLMs across diverse clinical scenarios.

However, existing benchmarks do not fully reflect the performance of LLMs in real-world clinical data, such as electronic health records (EHRs), which are critical as they document clinical status, history, and procedures that directly inform decision-making in clinical practice.¹⁸ The commonly-used benchmarks focus on medical questions sourced from medical examinations or derived from academic literature, exemplified by the United States Medical Licensing Examination (USMLE)^{19,20} or PubMed-based datasets.^{21,22} These benchmarks oversimplify LLM evaluation in clinical practice by relying on multiple-choice tasks and normalized text, failing to reflect the complexity of real-world data such as noisy notes, abbreviations, acronyms, and non-standard expressions.²³ As a result, LLMs like GPT-4²⁴ and Med-PaLM-2²⁵ achieve expert-level scores on the USMLE but perform unsatisfactorily in real-world clinical applications.^{26,27} The recently released HealthBench²⁸ simulates clinical scenarios without using real-world clinical text, and therefore does not reflect LLMs' capabilities in clinical text understanding. While some recent studies have evaluated LLMs in isolated

real clinical settings,^{29,30} they typically focused on narrow use cases and a limited set of LLMs,^{31,32} making it difficult to generalize findings to other clinical applications or track the state-of-the-art LLMs. In addition, limited research on multilingual medical benchmarks impedes the broader applicability of LLMs in global healthcare,^{33,34} raising concerns about the potential bias arising from underrepresented languages and regions.^{35–38} Such under-representative evaluations not only undermine the reliability and generalizability of LLMs but also amplify fairness and bias concerns, particularly across languages, regions, and healthcare systems.³⁹ When training and evaluation data are concentrated in high-resource languages or specific healthcare contexts, model outputs risk reinforcing existing inequities,⁴⁰ such as inferior performance in underrepresented populations or culturally specific clinical expressions.⁴¹

The rapid evolution of LLMs underscores a high demand for comprehensive and continuously updated medical leaderboards.⁴² With advanced technology and models emerging every few weeks, the landscape of LLMs is dynamic and ever-changing. Notably, the recent introduction of medical LLMs such as Med-PaLM1/2^{25,43}, MeLLaMA,⁴⁴ and Med-Found⁴⁵ also highlights the growing focus on improving performance and clinical relevance in healthcare. Leaderboards – objective displays of model performance across a wide variety of tasks – are essential for providing fair comparisons of LLM capabilities and tracking performance variations, thereby offering valuable guidance for subsequent model development and clinical implementation. Such leaderboards have already been widely implemented in non-medical fields⁴⁶ but have not yet been well adopted in clinical domains. As LLMs are progressively integrated and deployed into clinical practice, robust benchmarking is vital for proactively identifying and mitigating potential risks and biases before they impact patient care.⁴⁷ Therefore, establishing a unified, realistic, and multilingual clinical benchmark is crucial for bridging the gap between the theoretical capabilities of LLMs and their practical implementation in specific use cases and care settings.^{48,49}

To address the above challenges, we developed a large-scale and comprehensive benchmark that evaluates LLM performance on multilingual, real-world clinical text across diverse tasks. Building upon our systematic review of global clinical text resources,⁵⁰ this study proposes BRIDGE, a multilingual LLM benchmark that comprises 87 real-world clinical text tasks spanning nine languages

and more than one million samples. Reference standards for benchmark evaluation are sourced from the original data releases, including various forms of manual review and labels derived from structured data linked to the source EHRs.⁵⁰ To our knowledge, BRIDGE is the first comprehensive benchmark that focuses on real-world clinical data and is the largest benchmark for LLM in medicine to date. (A detailed comparison with existing benchmarks is provided in Supplementary Table S1.) We evaluated 95 advanced LLMs, including DeepSeek-R1,⁵¹ OpenAI GPT-4o,⁵² Google Gemini,^{53,54} Llama 4,⁵⁵ and Qwen3,⁵⁶ under three different commonly employed inference strategies (Zero-shot, Few-shot,⁵⁷ and Chain-of-Thought⁵⁸) through HIPAA-compliant environments. By integrating a systematic task taxonomy, we established a comprehensive leaderboard that not only provides a holistic perspective on LLM performance but also investigates their capabilities across various clinical settings, including inference strategies, languages, task types, and clinical specialties. This study provides critical insights and resources for integrating LLM into clinical practice, bridging the gap between LLM development and clinical applications.

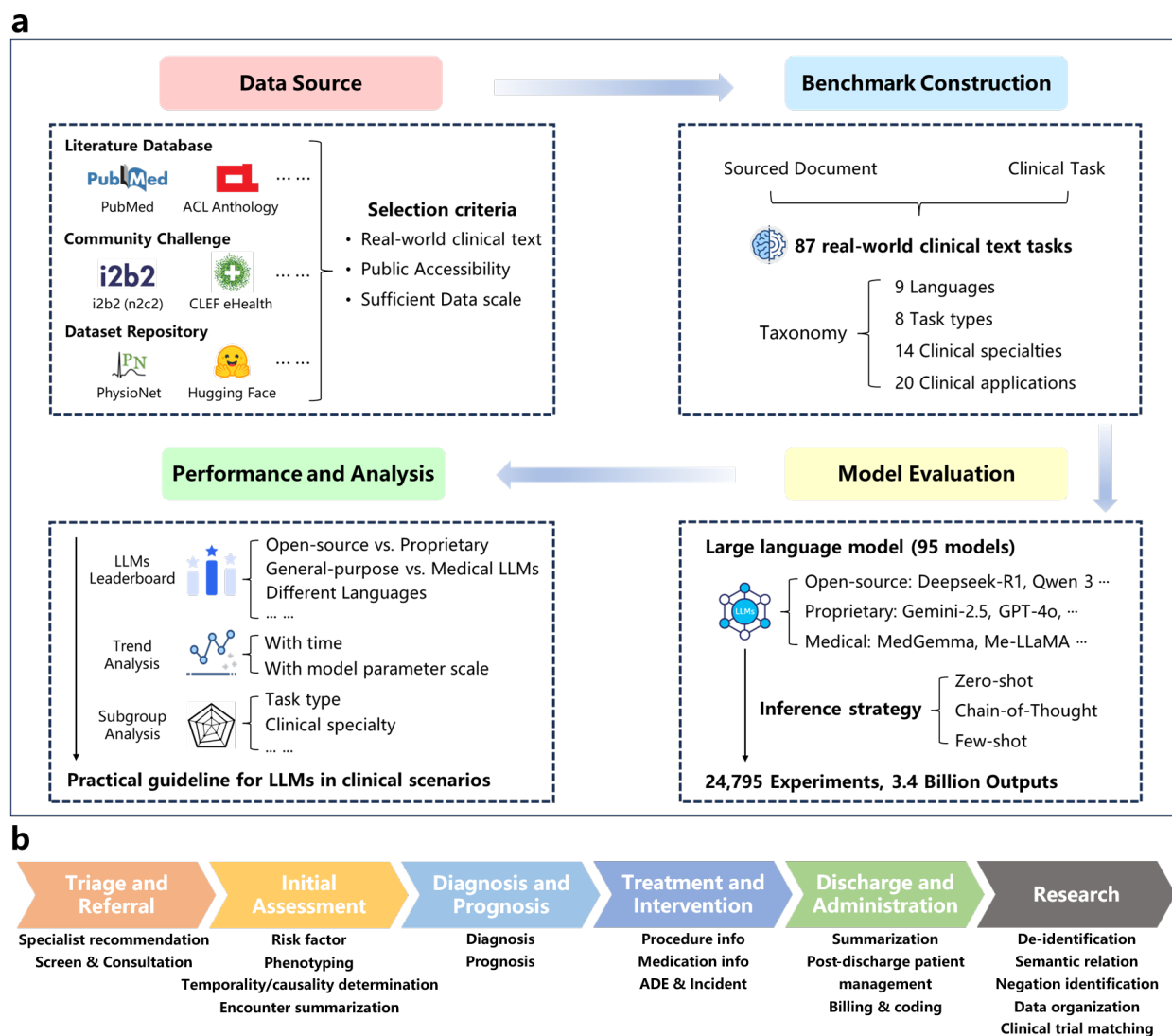


Figure 1. Overview of benchmarking large language models in clinical text understanding. (a) Workflow of benchmark construction, model evaluation, and performance analysis; (b) Clinical applications supported by the benchmark across different stages of patient care.

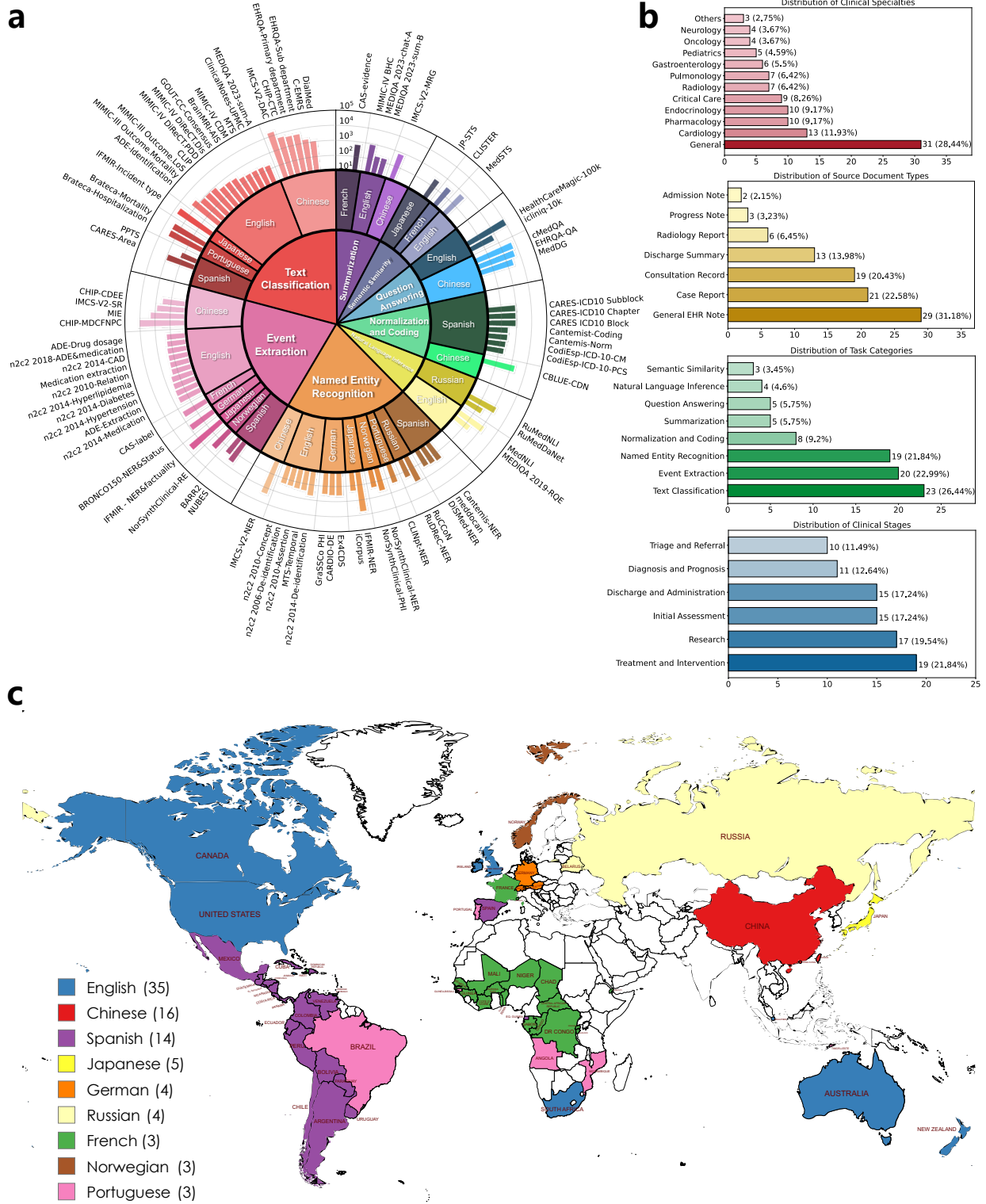


Figure 2. Overview of Benchmark Characteristics and Task Distribution. (a) Distribution of task types and associated languages, (b) Statistics on the distribution of clinical specialties, source document types, and task categories., and (c) Geographic distribution of countries where our benchmark covers the official languages.

Results

Benchmark Overview

The overall workflow of this study is illustrated in Figure 1. In total, our benchmark encompasses 87 tasks from 59 clinical text datasets. These tasks span nine languages and 1,418,042 samples, with 138,472 samples reserved for testing. Among them, 68 tasks (78.2%) are sourced from real-world EHR notes or clinical case reports, and 19 tasks (21.8%) are derived from real-world online patient-doctor consultation records. Figures 2a and 2b visualize the distribution of these tasks, covering 8 types, such as named entity recognition (e.g., phenotyping), classification (e.g., disease prediction), question answering, EHR summarization, and others. Figure 2c highlights this benchmark’s coverage of nine languages, distributed as follows: English (35 tasks, 40.2%), Chinese (16, 18.4%), Spanish (14, 16.1%), Japanese (5, 5.8%), German (4, 4.6%), Russian (4, 4.6%), French (3, 3.5%), Norwegian (3, 3.5%), and Portuguese (3, 3.5%). Detailed information about all tasks can be found in the Section Methods.

As shown in Figure 3a, we evaluated 95 state-of-the-art (SOTA) LLMs, covering proprietary, open-source models (0.6B to 671B parameters), and medically specialized models. All LLMs were assessed under three distinct inference strategies: Zero-shot, Chain-of-Thought (CoT), and Few-shot, totally 24,795 experiments. Comprehensive descriptions of the models, alongside their technical specifications, are available in Supplementary Table S2. Additionally, we investigated potential data contamination of the benchmark and found that most tasks did not appear to have been included in the training corpora of these models. Detailed analyses are presented in Supplementary Figure S1. The constructed benchmark and leaderboard are publicly available and will be regularly updated.

Overall Performance

We assessed LLMs using an overall score, defined as the average of their primary-metric scores across all tasks, to reflect their comparative performance across the entire benchmark, with the score ranging from 0 to 100 (Supplementary Table S3 for details). Figure 3b demonstrates the zero-shot performance of LLMs, while Figure 4 highlights the leading models under each inference strategy. Under the zero-shot setting, the three best-performing LLMs were Gemini-2.5-Flash⁵⁴ (44.8 [44.1-45.6], 95%CI), DeepSeek-R1⁵¹ (44.2 [43.5-45.0]), and GPT-4o⁵² (44.2 [43.4-45.0]). With CoT prompting, the overall

performance did not improve in general, with Gemini-2.5-flash (43.3 [42.5-44.1]) maintaining the top position, followed by Qwen3-Next-80B-Thinking (42.9 [42.1-43.7]) and DeepSeek-R1 (42.1 [41.3-42.9]). In contrast, few-shot prompting led to substantial performance gains, with Gemini-Series leading: Gemini-1.5-Pro (55.5 [54.7-56.3]), Gemini-2.5-Flash (53.4 [52.6-54.2]), and Gemini-2.0-Flash (53.3 [52.5-54.2]), followed by GPT-4o (52.6 [51.8-53.4]) and MedGemma-27b-it (52.0 [51.1-52.8]). Within the medical LLMs, these top-performing models showed distinct strengths across different inference strategies: HuatuoGPT-o1⁵⁹ achieved the highest score in the zero-shot setting (41.0 [40.2-41.8]), Baichuan-M2-32B⁶⁰ led under CoT (38.3 [37.4-39.2]), while MedGemma-27B-it⁶¹ excelled in few-shot (52.0 [51.1-52.8]). Overall, few-shot learning emerged as the most effective inference strategy for these clinical text tasks. Although LLM performance varied across different tasks, our leaderboard reveals the substantial ability gap of current LLMs for comprehensive clinical text understanding across diverse tasks, with the highest overall score on the whole BRIDGE only 55.5 under few-shot setting, indicating considerable space for further improvements.

LLMs Comparison

Figure 3b reveals a generally upward trajectory in overall performance under zero-shot setting across the evolving landscape of LLMs, illustrating both the rapid development and substantial potential of LLMs. While proprietary models, represented by Google Gemini series and OpenAI GPT-4o, maintain a performance lead, open-source LLMs have been rapidly advancing and narrowing the gap. The newly released DeepSeek-R1 (671B) and Qwen3-Next-80B-Thinking outperform several proprietary LLMs, signaling a shift toward increasingly competitive open-source performance. Other widely adopted open-source LLMs also achieved comparable performance, including Mistral-Large-Instruct (score of 42.3),⁶² Qwen2.5-72B-Instruct (41.6),⁶³ Gemma-3-27B-it (39.9),⁶⁴ and Llama-3.3-70B-Instruct (39.9)⁶⁵, as well as the derived variant Athene-V2-Chat⁶⁶ (41.7). Moreover, the progress across model generations highlights the rapid advance of LLM developments: the Qwen3 series consistently outperformed its Qwen2 predecessors, and Llama-3.3-70B-Instruct modestly surpassed Llama-3.1-70B-Instruct (39.9 [39.1-40.7] > 39.1 [38.3-39.9], $p = 0.16$). In contrast, the latest Llama-4-Scout-17B-16E-Instruct (109B)⁵⁵ showed a marked decline despite having more parameters (35.1 [34.4-35.9] vs. 39.9 [39.1-40.7], $p < 10^{-4}$).

Domain-specialized LLMs demonstrated promising yet inconsistent trends. For instance, MedGemma-27B-it outperformed the Gemma-3-27B-it (40.8 [40.0–41.6] > 39.9 [39.1–40.7], $p=0.12$). Nevertheless, most medical variants did not surpass their general-purpose counterparts: MeLLaMA-70B-chat (32.3) and Llama-3-70B-UltraMedical (33.4) scored lower than the related Llama-3.1-70B-Instruct (39.1) ($p<10^{-4}$). Some even lagged behind their foundation models — for instance, HuatuoGPT-o1-72B (41.0) trailed Qwen2.5-72B-Instruct (41.6), and Llama-3.1-8B-UltraMedical (20.2) fell well below Llama-3.1-8B-Instruct (29.0) ($p<10^{-4}$).

Model scale also emerged as a key determinant of performance. Figure 3b illustrates the performance gains associated with increasing model size, with larger models generally outperforming smaller ones. The comparisons between DeepSeek-R1 and its variants with different model sizes (Figure 3c) demonstrate a consistent improvement in performance as the size of model parameters increases, aligning with the trend observed in the Llama, Qwen, and MeLLaMA model families. Together, these results highlight the effectiveness of scaling laws in enhancing LLM performance for clinical applications.² Models within the 70-80B range represent the most common category of open-source LLMs and typically achieve robust performance, led by Qwen3-Next-80B-A3B-Thinking (43.9), Athene-V2-Chat (72B) (41.7), and Qwen2.5-72B (41.6). Among middle-sized LLMs (~30B parameters), Qwen3-30B-A3B-2507-Thinking (41.8), Qwen3-32B-Thinking (41.0), and MedGemma-27B-it (40.8) stand out.

Furthermore, recently developed reasoning LLMs exhibit distinct model behavior with notable performance, which enables explicit thinking processes during inference.^{6,51} As shown in Figure 3d, the LLM's performance on English text-classification tasks increases with longer CoT outputs, suggesting that in-depth thinking and reasoning may contribute to improved task accuracy. Representative reasoning models, such as DeepSeek-R1, gpt-oss-120B, and Qwen3-Next-80B-A3B-Thinking, provide substantially longer reasoning chains and outperform earlier non-thinking models. In parallel, emerging mixture-of-experts (MoE) architectures markedly improve computational efficiency by activating only a subset of experts during inference. For example, DeepSeek-R1 (671 B) activates approximately 37B parameters per forward pass, whereas Qwen3-Next-80B-A3B-Thinking

and Qwen3-30B-A3B-Thinking-2507 each activate about 3B parameters, achieving considerable efficiency gains without sacrificing accuracy.

Inference Strategy Performance

Figure 4a compares the performance of representative LLMs under three inference strategies: zero-shot, CoT, and five-shot (full comparison in Supplementary Table S3). Across the benchmark, few-shot prompting consistently yielded the highest accuracy, with 91 of 95 models (95.8%) achieving better results than in the zero-shot setting. Remarkably, 64 models (66.7%) showed performance gains exceeding 20%, underscoring the broad effectiveness of few-shot learning even when examples were randomly selected. Few-shot enhancements benefited both top-tier and lower-ranked LLMs. Among the leading models under zero-shot, DeepSeek-R1 improved from 44.2 to 51.4 (+7.2, +16.3%), while Gemini-1.5-Pro rose from 43.8 to 55.5 (+11.7, +26.7%), indicating that few-shot further augments even the most capable models. Similarly, models initially underperforming in zero-shot mode also exhibited significant improvements with few-shot prompting, such as the smaller LLMs (e.g., Llama-3.2-1B-Instruct from 12.7 to 24.4, +92.1%).

Notably, several medical LLMs benefited the most from the few-shot strategy. Llama3-OpenBioLLM-8B achieved the largest improvement, growing from 14.2 to 33.1 (+18.9, 133.1%), followed closely by Meditron-70B (15.7 to 32.1, +16.4, +104.5%). Other medical models, such as MedReason-8B, MMed-Llama-3,⁶⁷ HuatuoGPT-o1-8B, and Llama-3.1-8B-UltraMedical, also exhibited improvements ranging from 53.5% to 70.9%, indicating that task exemplars may be particularly valuable for domain-tuned models trained on limited instruction diversity. In contrast, explicitly enforcing step-by-step reasoning through CoT did not yield the expected performance gains for most LLMs, with only 11 models showing slight improvement.

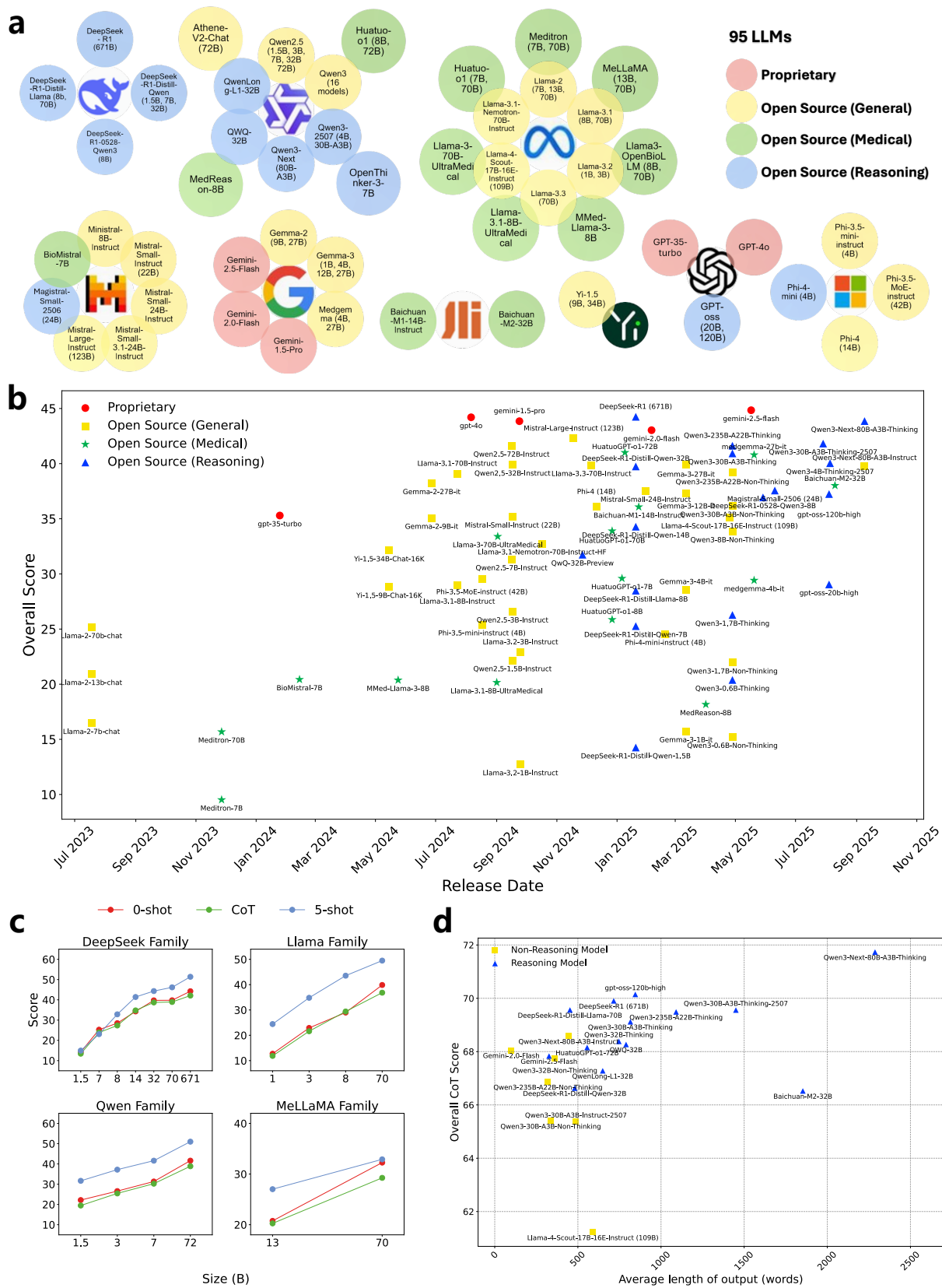
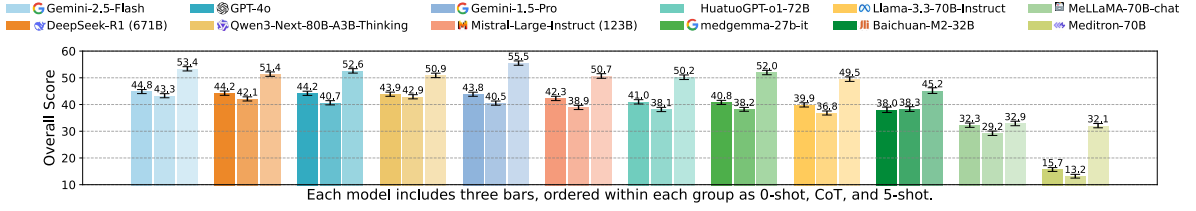


Figure 3. Overview of evaluated LLMs and their performance. (a) Categorization and information of evaluated LLMs. (b) Benchmark performance (zero-shot) of LLMs with their release dates. (c)

Comparative performance analysis of LLMs of varying sizes within the same model family. (d) Model ($\geq 30\text{B}$ and released after 2025/1/1) performance on English text classification tasks versus output length under CoT prompting.

a



b

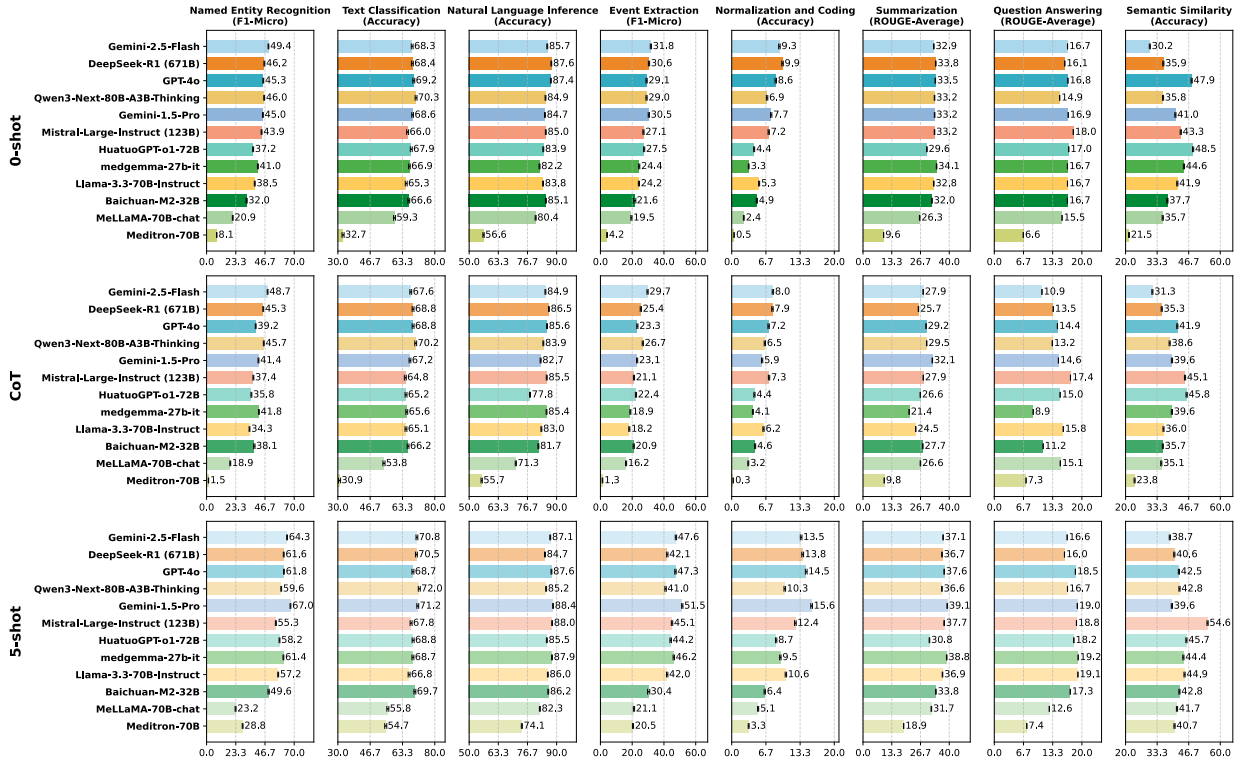


Figure 4. Comparative performance of 12 leading and commonly used LLMs under different inference strategies. (a) Overall score of LLMs evaluated across three inference strategies: zero-shot, CoT, and 5-shot prompting. For each model, three bars are displayed in the order of zero-shot, CoT, and 5-shot. (b) Performance of LLMs across different task categories under the three inference strategies. Error bars represent 95% CI.

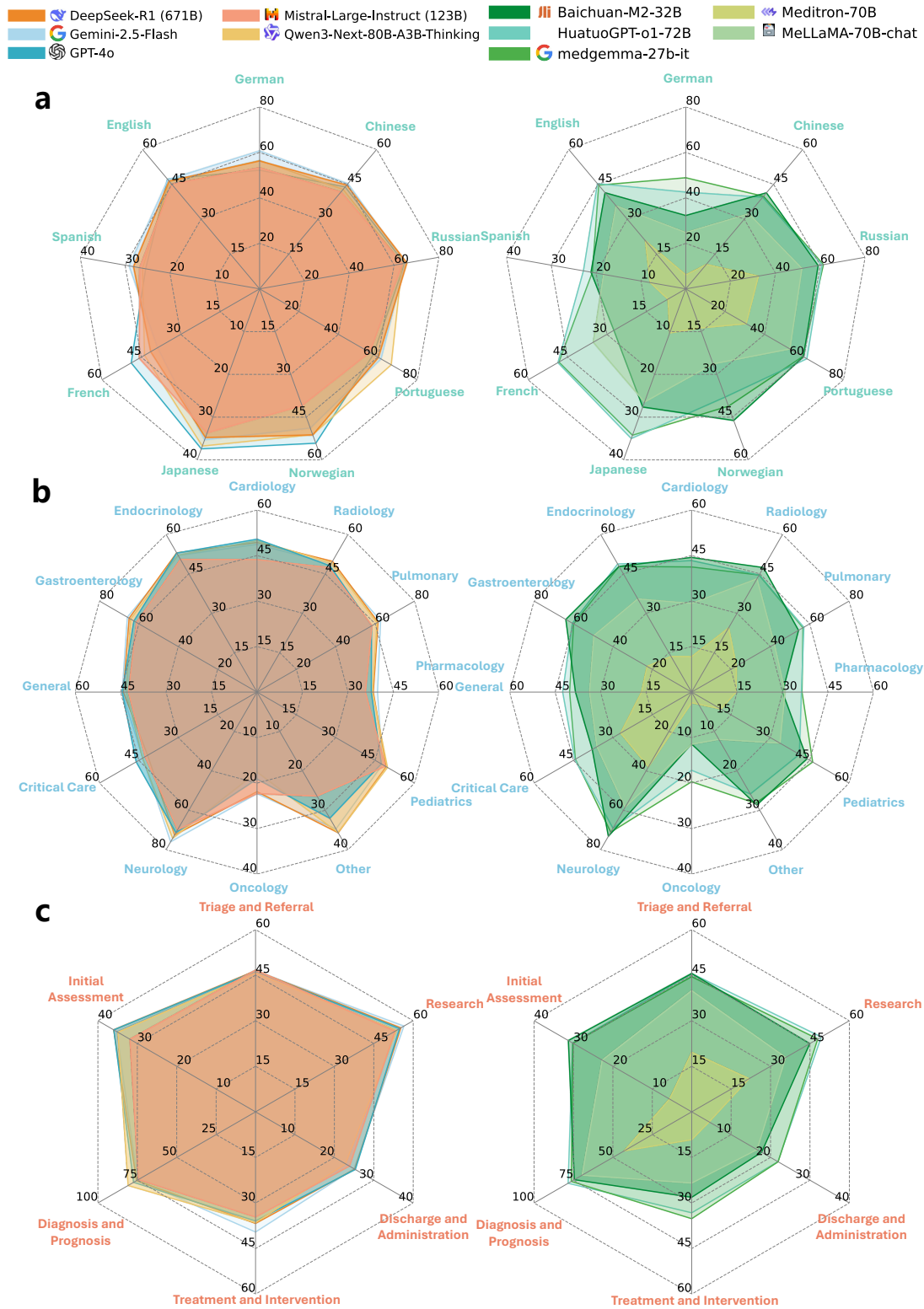


Figure 5. Zero-shot performance of the five leading and commonly used LLMs in both the general and medical domains across different BRIDGE subgroups. (a) languages, (b) clinical specialties, and (c) clinical stages.

Performance analysis for different task types

This benchmark encompasses a wide spectrum of clinical text tasks, and Figure 4b provides an overview of model abilities across different task types for both general and medical LLMs (Supplementary Table S4 for full details). Under zero-shot setting, gpt-oss-120B⁶⁸ achieved the highest accuracy in text classification (70.9), while Gemini-2.5-Flash excelled in information extraction, leading in both named entity recognition (NER, 49.4) and event extraction (31.8). DeepSeek-R1 attained top scores in natural language inference (NLI, 87.6) and normalization and coding (9.9), whereas HuatuoGPT-o1-72B showed the strongest performance in semantic similarity (48.5), Gemini-2.0-Flash performed best in question answering (QA, 18.4), and Gemma-2-27B-it led in summarization (34.7). In contrast, aside from HuatuoGPT-o1-72B and MedGemma-27B-it, most medically specialized LLMs struggled to generalize across diverse task types, with average ranks consistently below 20 out of 95 LLMs.

Figure 4b further illustrates the performance variations across different inference strategies. LLMs typically performed best on text classification and NLI, both of which offer well-defined, discrete label sets and thus present fewer ambiguities in the output. By contrast, information extraction tasks, such as NER and event extraction, benefited substantially from few-shot prompting, suggesting that more complex and context-dependent clinical tasks require examples to clarify detailed definitions and criteria. Meanwhile, normalization and coding tasks, which demand alignment with standardized medical coding systems (e.g., ICD-10), remained particularly challenging. Even with few-shot prompting, accuracy in these tasks remained relatively low (around 15%), reflecting the lack of embedded medical coding knowledge within most LLMs.²⁶ Similarly, text generation tasks, including QA and summarization, yield modest performance levels (around 20%), indicating that LLMs face significant challenges in clinical text generation.

Performance analysis for different languages

Figure 5a summarizes the performance of LLMs across different languages, with details provided in Supplementary Table S5. Since task composition varies by language, model performance was averaged across tasks within each language to indicate its language-specific ability. Overall, many advanced models exhibited robust cross-linguistic adaptability, maintaining consistent performance across

diverse languages. For instance, Gemini-2.5-Flash ranked first in Chinese, Spanish, and German, while Gemini-1.5-Pro performed best in English, and GPT-4o led in Japanese and Norwegian. Then, Qwen3-Next-80B-A3B-Thinking achieved the highest score in Portuguese, Qwen2.5-72B-Instruct led in French, and DeepSeek-R1 excelled in Russian. Based on the average score across languages, Qwen3-Next-80B-A3B-Thinking achieved the highest overall average (48.2). Notably, models from the Qwen family demonstrated exceptional multilingual competence, as exemplified by 12 of the top-20 models belonging to the Qwen2/3 series and their derived variants. This underscores the strong multilingual potential of well-optimized foundation models. However, for the parallel task of MedNLI (English) and RuMedNLI (Russian), all LLMs achieved better performance on MedNLI, indicating a consistent performance degradation in non-English clinical contexts.

Among specialized medical LLMs, HuatuoGPT-o1-72B exhibited the broadest cross-linguistic versatility, achieving top performance in five languages (English, Spanish, French, Japanese, and Portuguese). MedGemma-27B-it showed competitive results in German and Russian, while Baichuan-M2-32B performed best in Chinese and German. However, English-centric medical LLMs (e.g., MeLLaMA and BioMistral) perform comparatively lower when applied to other languages. These results emphasize the necessity for more diverse multilingual corpora and language-specific tuning to ensure effective global deployment of LLM-based solutions.

Performance analysis for different clinical specialties and clinical stages

We further examined model performance across various clinical specialties (Figure 5b), reflecting the varied medical background from which datasets originated or the specific clinical challenges they addressed (Supplementary Table S6). As Figure 2b shows, this benchmark comprises 14 specialties, with “General” denoting datasets that span more than five domains or unspecified contexts, and “Others” including nephrology, dermatology, and psychology (one dataset each). Across specialties, Gemini-2.5-Flash consistently delivered top performance in gastroenterology (65.8), neurology (76.0), pulmonology (62.8), pharmacology (39.7), and general datasets (44.5). DeepSeek-R1 achieved the highest scores in radiology (49.8) and the “Others” category (35.7). Gemini-1.5-Pro led in critical care (46.6), pediatrics (50.4), and oncology (22.5), while GPT-4o excelled in cardiology, and Qwen3-30B-A3B-Thinking-2507 performed best in endocrinology. Despite domain-focused pretraining and

supervised fine-tuning, the specialized medical LLMs did not show superiority over their general-purpose counterparts. Among these, MedGemma-27B-it demonstrated the broadest versatility, ranking first in 3 clinical specialties and achieving the highest average score across specialties (44.7), placing 14th among all 95 models—notably surpassing several larger general models such as Qwen3-Next-80B-A3B-Instruct and DeepSeek-R1-Distill-Llama-70B. These findings suggest that well-targeted domain adaptation can confer meaningful advantages even without further scaling.

Figure 5c further presents LLM performance across different clinical stages (Supplementary Table S7). The Gemini series demonstrated strong results: Gemini-1.5-Pro achieved the best performance in Initial assessment (37.1), Gemini-2.0-Flash excelled in Triage and Referral (47.5), and Gemini-2.5-Flash outperformed others in both Treatment and Intervention and Research-related tasks. GPT-4o and gpt-oss-120B led in Discharge and Administration, while among medical models, MedGemma-27B-it, HuatuoGPT-o1-72B, and Baichuan-M2-32B each delivered top performance in two clinical stages.

Overall, LLMs demonstrated stronger performance in the Diagnosis and Prognosis stage, likely due to the prevalence of well-structured classification tasks. In contrast, other stages often involve more complex tasks such as information extraction (e.g., phenotyping, temporal or causal relations) and text generation (e.g., summarization), which may pose greater challenges.

Discussion

This study represented the largest benchmark to date for LLMs evaluations on multilingual, real-world clinical text, encompassing 87 tasks in nine languages. We developed a systematic framework and leaderboard for categorizing tasks and defining corresponding evaluation methods, enabling a thorough assessment of 95 state-of-the-art LLMs with 24,795 experiments and 3.43 billion inferences in total. Beyond providing a holistic view of current LLM capabilities, our analyses examined performance variations across different perspectives, including inference strategies, languages, task types, and clinical specialties. With comprehensive analyses and a continuously updated leaderboard, clinicians can employ this benchmark to determine the candidate LLMs that best fit specific clinical or research contexts and deployment environments, while AI developers in healthcare can leverage it as a robust reference for further model fine-tuning and system integration. For patients, the benchmark

serves as a preliminary assessment of the reliability of LLM outputs, thereby promoting better transparency and confidence in AI-assisted healthcare services.

Unlike scientific literature or licensing exam questions, the information in our benchmark is drawn from actual patient care – the EHR and online doctor-patient interactions.^{69,70} Differing from the simplified and standardized multiple-choice evaluations,¹⁸ BRIDGE includes tasks specific to the administration and provision of health care, better reflecting the multifaceted capabilities of LLMs. Although certain LLMs achieved scores above 80 on standardized exams^{25,32}—for example, Deepseek-R1 reached a score of 92 on the USMLE dataset,³² it attained an overall score of only 44.2 out of 100 on our benchmark. While this was the highest among the models evaluated, there remains substantial room for improvement. This stark discrepancy is sobering and highlights limitations of the current LLMs in clinical applications and the necessity of more clinically oriented evaluation before integrating LLMs into clinical practice.⁷¹ Furthermore, many existing models and benchmarks remain English-centric, which restricts both the generalizability and equitable deployment of LLMs in global healthcare. This underrepresented situation not only constrains accessibility but also introduces systematic biases and fairness, leading to performance degradation and even contextually inappropriate or unsafe clinical interpretations when applied to non-English settings.^{36,39,72} Our results also reveal substantial cross-linguistic variation and persistent challenges in multilingual medical reasoning. In addition, BRIDGE captures a broad spectrum of real-world clinical contexts, from early-stage triage and referral to diagnosis, treatment recommendation, and billing coding, thereby avoiding evaluations that overemphasize any single dimension.⁶⁷ Collectively, by encompassing nine languages and diverse applications, BRIDGE facilitates more inclusive, globally applicable, and clinically relevant evaluation in medical AI, extending the benefit of LLMs to diverse healthcare systems worldwide.⁷³

Given the rapid and transformative progress of LLMs,⁷⁴ especially fueled by open-source initiatives such as Qwen, LLaMA, and DeepSeek,^{63,65} our evaluation of 95 cutting-edge LLMs provides timely and valuable insights on integrating LLMs into clinical applications. The results reveal the continuous progress of open-source models, exemplified by Deepseek-R1 surpassing several evaluated proprietary systems (e.g., Gemini-2.0-Flash).⁷⁵ These open-source foundation models also expedited the development of medical LLMs, with mid-sized domain models such as MedGemma-

27B and Baichuan-M2-32B showing encouraging performance, suggesting the potential of targeted domain adaptation. Nevertheless, most medical-specific variants generally underperform their general-purpose counterparts. This gap partially stems from the outdated base models these medical LLMs were built on, and the limited clinical relevance of their training data,⁷⁶ which is mainly drawn from biomedical literature (e.g., PubMed) rather than EHR data.^{77,78} Although some models (e.g., BioMistral-7B and MMed-Llama 3) have undergone domain-specific pretraining and demonstrate strong potential, they often lack broad multi-task instruction fine-tuning or disproportionately rely on medical question datasets. The lack of task diversity may lead to overfitting and reduce the LLM's generalizability across tasks.⁷⁹ This limitation is reflected in their weaker performance under zero-shot settings and the substantial gains from few-shot prompts, which are compensated for by injecting task-specific context.⁸⁰ Moreover, models' ability to accurately interpret task instructions and generate valid responses (as shown in Supplementary Table S8) varies considerably; several smaller and medical-specific LLMs exhibit lower valid-response rates, which further undermine model performance. Additionally, models such as HuatuoGPT-o1-72B, which incorporated reinforcement learning on top of an instruction-tuned base (i.e., Qwen2.5-72B-Instruct), demonstrated notable performance gains with robust versatility to broad tasks.⁵⁹ Despite the multilingual capabilities inherited from extensive web-data pretraining,^{34,81} most medical LLMs remain primarily optimized for English, leaving genuinely multilingual clinical foundation models relatively underexplored.⁸² Additionally, open-source solutions facilitate on-site deployments within hospitals, enabling more localized data and model governance, thereby reducing potential privacy and security risks.⁸³ Our experiments further confirm that scaling laws² persist within clinical tasks: the larger models significantly and consistently outperform the small ones, which emphasizes the enduring trade-off between performance and computational cost.

Inference strategy plays a pivotal role in the practical deployment of LLMs, particularly since most clinical applications will likely not involve LLM fine-tuning but rather rely on task-specific prompting.⁸⁴ Our findings indicate that few-shot prompting proves highly effective for clinical text tasks, significantly enhancing task-specific comprehension and contextualization with only five randomly selected examples. To enhance interpretability, CoT explicitly instructs LLMs to generate

step-by-step reasoning before arriving at a final answer.⁵⁸ Contrary to observations in other domains,⁸⁵ CoT did not yield consistent performance gains in our benchmark and mostly impaired results.²⁰ This discrepancy appears to stem from the lack of sufficiently grounded medical knowledge to support accurate multi-step reasoning and the heightened risk of hallucinations in such a knowledge-intensive setting.^{86,87} Meanwhile, the newly developed reasoning LLMs (e.g., DeepSeek-R1 and Qwen3-Series), which employ reinforcement learning to strengthen reasoning capabilities, achieved superior results on our benchmark.^{51,88} These models demonstrated a promising direction for developing interpretable LLMs that more closely mirror human decision-making processes. Beyond improved reasoning, the integration of external domain knowledge, such as via retrieval-augmented generation,⁸⁹ can further enhance LLM performance and reliability, leveraging their in-context learning abilities while mitigating the risk of misinformation in clinical applications.⁸⁹

Because of the tremendous potential of LLMs in healthcare scenarios as well as the considerable risks to patient safety, transparent and robust benchmarking will be essential to adopt LLMs into clinical practice.^{90,91} Future benchmarks should encourage closer collaboration among clinicians, patients, and LLMs to better simulate real-world interactions, enhance overall clinical impact, and provide a validated human baseline.⁹² Meanwhile, adopting standardized reporting frameworks (e.g., TRIPOD-LLM⁹³) are essential for enhancing transparency, reproducibility, and comparability in future medical applications, thereby harmonizing evaluation and broad clinical practices. Furthermore, LLMs have been observed to exhibit overoptimism in their inferences,⁹⁴ which is critical to address in medical applications. Our benchmark provides a foundation for evaluating the trustworthiness of such predictions. Given the complexity of clinical practice, the scores in our benchmark do not fully equate to LLM performance on specific clinical applications, which require further rigorous assessment. However, BRIDGE provides timely and comprehensive comparisons across diverse tasks and models, serving as a valuable starting point for model selection and filtering, guiding decisions before committing to resource-intensive evaluations or further development of selected base models.

This study has several limitations. First, given the large scale of the dataset, we used reference labels obtained from the original data sources rather than labels verified by human clinicians. Second, as concerns regarding privacy and safety restricted the availability of accessible clinical text datasets,

certain tasks may rely on overlapping data sources, restricting corpus diversity. Third, this work primarily evaluated LLMs under common inference settings without incorporating specific instruction optimization. We will further examine more strategies (e.g., instruction tuning, fine-tuning, and RAG) to enhance model adaptability to clinical workflows in future studies. Fourth, the overall score was calculated by averaging the primary metrics across task types, which may introduce inconsistencies due to varying metrics. However, we provide fine-grained comparisons on the leaderboard across task types under the same metric to ensure fair comparisons. Finally, while we investigated 95 advanced models, several newly released LLMs (e.g., OpenAI o3, Gemini-Pro-2.5-Pro, and Med-PaLM2) were not evaluated due to the constraints of model access and resource limitations within HIPAA-compliant healthcare systems. Nonetheless, this leaderboard is continuously updated to maintain its relevance and currency, offering a dynamic resource for tracking and advancing LLM performance in clinical text understanding.

In conclusion, this study established a comprehensive benchmark and systematically evaluated LLMs on real-world clinical text understanding. By centering on real-world EHR-based tasks and capturing the complexity of clinical text, our findings highlight the gap between current LLM capabilities and the demands of clinical practice, while also revealing substantial performance variability across models, languages, and clinical scenarios. These insights provide critical guidance for optimizing LLMs in healthcare and inform future efforts to align model development with the practical needs of clinical applications.

Methods

Clinical Text Dataset Collection

To comprehensively evaluate the LLMs performance on real-world clinical text data, we systematically identified and curated a diverse collection of clinical text datasets representative of authentic clinical scenarios.

This process was initially guided by our prior systematic review of clinical text datasets,⁵⁰ which conducted a global survey of publicly available resources. Building upon this foundation, we expanded our search scope and employed a standardized protocol to ensure that the included datasets fully satisfied the benchmarking criteria of clinical relevance, diversity, and suitability.

Specifically, we targeted three primary sources:

1. **Literature Databases:** Widely recognized biomedical literature databases, including PubMed and MEDLINE, and computational linguistic repositories, notably the ACL Anthology, a leading digital archive of Natural Language Processing (NLP)-focused journal articles and conference proceedings.
2. **Community Challenges:** Commonly used and actively maintained clinical NLP challenges and benchmarks, such as the National NLP Clinical Challenges⁹⁵ (n2c2, formerly i2b2) and CLEF eHealth.⁹⁶
3. **Dataset Repositories:** Biomedical dataset platforms (PhysioNet⁹⁷) and NLP-focused dataset hubs (Hugging Face⁹⁸), which store extensive collections of biomedical datasets and are frequently updated with new resources.

Detailed search strategies, including specific Medical Subject Headings (MeSH) terms and keywords, can be found in our prior review.⁵⁰

Criteria for Dataset Selection

Datasets identified through these sources underwent screening based on the following predefined inclusion and exclusion criteria:

1. **Real-World Clinical Text Data:** Eligible datasets were required to consist of authentic clinical texts derived directly from real-world medical settings, such as electronic health

records (EHRs), clinical case reports, or healthcare consultations. Non-clinical sources (e.g., textbooks, social media) or datasets relying primarily on non-textual (e.g., genomic or protein sequences) or multimodal inputs were excluded.

2. **Public Accessibility and Availability:** Datasets included in this investigation are publicly available or accessible through standardized request procedures to ensure reproducibility and transparency.
3. **Sufficient Data Scale:** Only datasets containing at least 200 samples were included to ensure reliable evaluations and robust statistical analyses.

Finally, the curated datasets provided a diverse corpus representative of authentic clinical scenarios.

Benchmark Construction

Based on the included datasets, we constructed a set of clinical text tasks tailored for assessing LLMs. These tasks simulate diverse clinical scenarios characterized by complexity, contextual variability, and multi-source information requirements. Unlike traditional NLP methods, which typically rely on supervised training with task-specific model architectures, LLMs perform tasks by interpreting textual prompts without dedicated training. Therefore, precise task design and standardization are critical for fair and objective evaluations. Detailed information about all tasks can be found in Supplementary Section 4 and Section 5, and the metadata of tasks are in the Supplementary Table S9.

To ensure task suitability and consistency, we transformed and standardized datasets through the following structured process:

Task Definition and Categorization

Task objectives and evaluation criteria were primarily derived from original dataset descriptions or primary publications (hereafter collectively referred to as dataset source). Tasks were categorized into different types:

1. Text classification: Determine or predict categorical labels (e.g., diagnosis, risk stratification) based on the provided clinical texts.
2. Semantic similarity: Assessing the similarity of two sentences or clinical notes.
3. Natural Language Inference (NLI): Evaluating the logical relationships (e.g., entailment, contradiction, neutrality) between paired texts.

4. Normalization and coding: Map the whole clinical note or the extracted entities to standardized clinical code systems (e.g., ICD, SNOMED)
5. Named Entity Recognition (NER): Identify the medical entities and label them with appropriate types (e.g., symptom, disease, examination).
6. Event extraction: Identify the medical entities and capture additional attributes or relations beyond simple entity types (e.g., temporal status, severity).
7. Question-Answering (QA): Generating accurate responses to healthcare inquiries.
8. Summarization: Condense clinical notes into concise summaries by retaining essential information, with extraction or generation methods.

Input Text Preparation and Standardization

Relevant textual information for each task was systematically extracted from the original datasets and integrated using standardized templates. For instance, the required text fields were distilled from the whole EHR database and then condensed into structured inputs with templates (e.g., "Chief Complaint: ..., Examination: ..."). Additionally, for tasks introducing structured metadata (e.g., demographic information and examination results), we transformed these structured features into explicitly labeled textual forms, integrating them seamlessly with clinical notes. The template for input and output can be found in Supplementary Section 4.1.

Output Standardization and Formatting

Given the heterogeneity of original task outputs, including classification logits, BIO-style entity tags, or structured annotations, all outputs were instructed to be standardized into clear, structured textual responses for automatic result processing and analysis on evaluation. Specifically:

1. Tasks for Text classification, Semantic similarity, NLI, and Normalization and Coding (document level): Outputs were standardized into explicit textual labels indicating predicted categories.
2. Tasks for NER, Event extraction, and Normalization and Coding (entity level): Outputs were formatted into structured textual annotations clearly indicating subject spans and other required attributes (e.g., "Entity: ..., Type: ..., Status: ...").

3. Tasks for QA and Summarization: Outputs were directly formulated as concise, structured free-text outputs.

Outputs that failed to follow the required format were regarded as invalid responses during the evaluation phase. This uniform output format enables automated extraction and quantitative evaluation of LLM-generated results, ensuring efficient and objective performance assessment.

Task Instruction (Prompt) Definition

Due to the complexity of clinical text tasks, which often rely on professional definitions and domain-specific terminology, we prioritized the use of task instructions from authoritative dataset sources, including original dataset papers, annotation guidelines, and supplementary data descriptions. To minimize variability and potential biases introduced by differing prompts, we adopted straightforward and concise instruction templates for all tasks.

Given the multilingual nature of our benchmark, we preferentially retained the task description aligned with their original language if available, preserving context-specific semantics critical for accurate interpretation. Meanwhile, the instructions for other tasks and the base templates (both input and output may involve) were uniformly provided in English, as the existing LLMs all support English and yield fine performance in English.

Dataset Splitting

Dataset partitions followed the official splits defined by dataset sources whenever available, facilitating direct comparability with prior studies. For datasets without predefined splits, we applied the following selection strategy: for datasets with over 2000 samples, 10% were randomly selected as the test set; for datasets with 1000 to 2000 samples, 20% were selected; and for datasets with fewer than 1000 samples, all samples were used for testing except for 20 cases reserved as a pool for selecting few-shot examples. For each dataset, five samples outside the test set were randomly selected as few-shot examples. All the benchmark experiments were conducted on the split test partitions.

Task Taxonomy and Characteristics

To systematically investigate the abilities of LLMs across different clinical scenarios, we extracted key features for each task and mapped them into standardized taxonomies. These include

Language, Sourced Clinical Document, Clinical Specialty, and Clinical Stage and Application, which can refer to Supplementary Section 4.2 Task Taxonomy and Characteristics.

Model Implementation

We included a diverse range of state-of-the-art LLMs, covering both proprietary and open-source LLMs. Detailed information for all models is provided in Supplementary Table S2. All experiments, including data selection and model inference, used a fixed random seed (42) across all tasks and models to ensure reproducibility. For decoding configuration, unless explicitly specified by the model's official documentation, all models were evaluated using a greedy decoding strategy (temperature = 0, top_p = None, top_k = None) to eliminate randomness and produce deterministic outputs. For the Qwen3 series, the decoding parameters recommended in its official guideline were adopted.

The model inference was conducted using the following computational setups:

1. Open-source models: all open-sourced models (except DeepSeek-R1[671B]) were deployed locally on Mass General Brigham (MGB) institutional server with 8*NVIDIA H100 GPUs. The inference process was accelerated using the vLLM framework⁹⁹ to optimize efficiency. DeepSeek-R1 was deployed on Microsoft Azure managed by Stanford University.
2. Proprietary models: Due to privacy and security considerations, proprietary models were evaluated via institutional cloud infrastructure within the compliance of HIPAA:
 - a) OpenAI models (GPT-35-Turbo-0125 and GPT-4o-0806): Deployed on Microsoft Azure server at Mass General Brigham.
 - b) Google models (Gemini series): Deployed on Google Cloud Platform at Mayo Clinic.

All records of inference requests and responses were securely stored on institutional servers at MGB in accordance with MGB data governance policies. All experiments were conducted in full compliance with the HIPAA regulations.

Inference Strategy

In this study, we systematically evaluated three distinct inference strategies:

1. Zero-shot: Only the task instructions and input data were provided. The LLM was prompted to directly produce the target outputs without any support.

2. Chain-of-Thought (CoT): Task instructions explicitly directed the LLM to generate a step-by-step explanation of its reasoning process before providing the final output, which can significantly improve the model's interpretability.
3. Few-shot: Five reserved independent samples serve as examples, which leverage the LLM's capability of in-context learning to guide the model to conduct tasks. For models supporting conversational interactions, examples were presented sequentially to simulate realistic user-system dialogues; otherwise, input-output pairs were directly appended to the instruction.

Details about the prompt for different inference strategies can be found in Supplementary Section 4.1.

Evaluation Framework

To ensure consistent, objective, and automated evaluations, we established standardized evaluation schemes covering reference standard, result extraction, task-specific metrics, and statistical analysis.

Reference Standard

For each task in our benchmark, the reference standard is sourced from the labels released with the original source datasets. These labels were generated through different mechanisms, including expert manual annotation and derivation from structured EHR systems, and underwent the quality-control procedures defined by dataset creators. To maintain consistency with prior work and preserve the integrity of each dataset, we adopted these original labels without additional modification.

Result Extraction

We develop an automated script for each task separately to extract results from the standardized LLM outputs described previously. For outputs failing to meet the required formatting standards, we regarded them as invalid responses. We calculated the valid rate for each experiment setting and presented the results in Supplementary Table S8. For tasks under the types of text classification, semantic similarity, NLI, and normalization/coding (with explicitly defined labels), invalid model outputs were replaced with randomly assigned labels from the valid label set. For the remaining task types, invalid outputs were retained as empty responses.

Evaluation Metrics

Representative metrics were carefully selected for each task category, with a designated primary metric facilitating overall benchmarking comparisons:

1. Text classification, Semantic similarity, NLI, and Normalization and Coding (document level): We evaluate these tasks with Accuracy (primary metric), micro F1-score, and macro F1-score. Accuracy directly reflects overall classification performance by measuring the proportion of correct predictions. The F1 scores provide complementary insights by considering precision and recall across classes. The micro F1-score emphasizes performance in common classes, while the macro F1-score equally weights all classes, highlighting performance in less frequent categories.
2. NER, Event extraction, and Normalization and Coding (entity level): We evaluate these tasks with subject-level F1-score and event-level F1-score (both calculated by micro-scoring). The subject-level F1-score only evaluates the model's ability to identify the correct subjects without considering their attributes, providing preliminary performance insights. Event-level F1-score, the primary metric, comprehensively evaluates model accuracy by measuring extraction precision and recall across entities and their attributes.
3. QA and Summarization: We evaluate these tasks with BLEU-4, ROUGE-average (primary metric), and BERTScore. ROUGE-average¹⁰⁰ is the average score of ROUGE-1, ROUGE-2, and ROUGE-L, thus capturing the recall of unigrams, bigrams, and longest common subsequences between the candidate and reference texts. BLEU-4¹⁰¹ combines the precision of 1-, 2-, 3-, and 4-gram matches between generated outputs and references. BERTScore¹⁰² evaluates semantic alignment between generated and reference texts by leveraging contextual embeddings from BERT. All these metrics generally show a consistent trend across tasks, while the ROUGE-average exhibits more distinctions among models in our experiments. Therefore, we adopt ROUGE-average as the primary metric.

These metrics were computed using standard libraries to ensure reproducibility: *Scikit-learn*¹⁰³ for classification and extraction tasks, *nltk*¹⁰⁴ for BLEU-4, *rouge_scorer*¹⁰⁵ for ROUGE-average, and *bert_score*¹⁰⁶ for BERTScore.

Performance Calculation

To enable quantitative comparisons, we compute an overall score for each LLM by averaging its primary-metric values across all tasks. This aggregate measure reflects a model's relative performance

on the benchmark as a whole. For subgroup analyses, such as task type, language, clinical specialty, and clinical stages, the same averaging procedure is applied to the subset of tasks that meet the specified criteria, yielding a focused performance estimate within that domain.

Statistical Analysis

Model performance was evaluated with non-parametric bootstrapping (1,000 resamples with replacement). For each model, we computed the bootstrapped mean and its 95 % confidence interval (CI), yielding robust estimates of central tendency and sampling variability. Pairwise model comparisons were assessed with a two-sided significance test. All analyses were performed in Python 3.10.15 (NumPy v1.26.4, SciPy v1.14.1).

Data Contamination Analysis

The advanced LLMs typically undergo extensive training on vast data, raising the possibility of unintended data exposure of benchmark. To assess this potential data contamination, we employed a text completion-based approach to detect possible leakage of benchmark data into the evaluated models’ training corpora.¹⁰⁷ Specifically, we tokenized each test sample using model-specific tokenizers, truncating sequences at predetermined positions (tokens 10, 15, 20, 25, and 30) and prompting the LLM to predict the subsequent five tokens. The accuracy of predicted tokens compared to actual tokens (5-gram accuracy) was measured. A test sample was classified as potentially leaked if predictions exactly matched actual tokens at three or more truncation positions. To balance sufficient contextual information while avoiding overly simplistic long-context completions, truncation points began from token position 10, incrementing by intervals of 5 tokens.

Acknowledgments

We thank Xiaocong Liu, Wanxin Li, Qingcheng Zeng, Zichang Su, and Xiaoyue Wang for the initial collection and process of datasets. This study was partially funded by PCORI ME-2022C1-25646, Goldberg Scholarship and Brigham Research Institute. L.A.C. is funded by the National Institute of Health through DS-I Africa U54 TW012043-01 and Bridge2AI OT2OD032701, the National Science Foundation through ITEST #2148451, and a grant of the Korea Health Technology R&D Project through the Korea Health Industry Development Institute (KHIDI), funded by the Ministry of Health & Welfare, Republic of Korea (grant number: RS-2024-00403047). S.S. is funded in part by FDA research contracts (HHSF223201710186C and HHSF223201710146C), the FDA Sentinel Innovation Center (75F40119F19002), the NIH (NHLBI R01-HL141505, NIAMS R01-AR080194), the Burroughs Wellcome Fund, and PCORI. J.H.C. was supported by NIH/National Institute of Allergy and Infectious Diseases (1R01AI17812101), NIH-NCATS-Clinical & Translational Science Award (UM1TR004921), NIH/National Institute on Drug Abuse Clinical Trials Network (UG1DA015815 - CTN-0136), Stanford Bio-X Interdisciplinary Initiatives Seed Grants Program (IIP) [R12] [JHC], NIH/Center for Undiagnosed Diseases at Stanford (U01 NS134358), Stanford Institute for Human-Centered Artificial Intelligence (HAI), and Gordon and Betty Moore Foundation (Grant #12409).

Author contribution

J.Y. designed this study. J.W. and B.G. constructed the benchmark and conducted the experiments. J.W., B.G., and J.Y. analyzed the results and drafted the initial manuscript. R.Z. and K.X. contributed to the benchmark construction, J.W., B.G., and R.Z. contributed to the data contamination experiments, and K.X. and J.W. contributed to building the leaderboard. D.S., V.C., S.R-B, and Y.J. contributed the inferences of Gemini-series and Deepseek-R1. R.W., R.J.D., E.A., L.A.C., A.R., S.S., J.H.C., S.R-B., K.J.L., and J.Y. contributed to manuscript design and refinement. J.Y. and K.J.L. supervised this study. All authors revised, read, and approved the manuscript.

Competing interests

K.J.L. has received research grants from Takeda, AbbVie, and UCB for projects unrelated to this study. E.A. reports consultant fees from Fourier Health. S.S. is participating in investigator-initiated grants to the Brigham and Women’s Hospital from Boehringer Ingelheim, Takeda, and UCB unrelated to the topic of this study. He is an advisor to and owns equity in Aetion Inc., a software manufacturer. S.S. is an advisor to Temedica GmbH, a patient-oriented data generation company and his interests were declared, reviewed, and approved by the Brigham and Women’s Hospital in accordance with their institutional compliance policies. J.H.C. reports cofounding Reaction Explorer, that develops and licenses organic chemistry education software, and receive medical expert witness fees from Sutton Pierce, Younker Hyde MacFarlane, Sykes McAllister, Elite Expert, consulting fees from ISHI Health, and honoraria or travel expenses for invited presentations by insitro, General Reinsurance Corporation, Cozeva, and other industry conferences, academic institutions, and health systems.

Data availability

All fully open-access datasets in BRIDGE are shared at <https://huggingface.co/datasets/YLab-Open/BRIDGE-Open> under their respective data use agreements. All dataset sources and corresponding links for additional data access are listed in Supplementary Section 4. The leaderboard is available at <https://huggingface.co/spaces/YLab-Open/BRIDGE-Medical-Leaderboard>, where we also provide contact information for submitting new models for evaluation without requiring direct data access. Additional metadata supporting this study are available from the corresponding author upon reasonable request.

Code availability

The corresponding benchmark evaluation code can be found at <https://github.com/YLab-Open/BRIDGE>.

References

1. Thirunavukarasu, A. J. *et al.* Large language models in medicine. *Nat. Med.* **29**, 1930–1940 (2023).
2. Kaplan, J. *et al.* Scaling Laws for Neural Language Models. Preprint at <http://arxiv.org/abs/2001.08361> (2020).
3. Zhang, S. *et al.* Instruction tuning for large language models: A survey. (2024).
4. Min, B. *et al.* Recent advances in natural language processing via large pre-trained language models: A survey. *Acm Comput. Surv.* **56**, (2023).
5. Qin, C. *et al.* Is ChatGPT a general-purpose natural language processing task solver? in *Proceedings of the 2023 conference on empirical methods in natural language processing* 1339–1384 (Association for Computational Linguistics, Singapore, 2023). doi:10.18653/v1/2023.emnlp-main.85.
6. Huang, J. & Chang, K. C.-C. Towards reasoning in large language models: a survey. in *Findings of the association for computational linguistics: ACL 2023* 1049–1065 (Association for Computational Linguistics, Toronto, Canada, 2023). doi:10.18653/v1/2023.findings-acl.67.
7. Tang, L. *et al.* Evaluating large language models on medical evidence summarization. *Npj Digit. Med.* **6**, 1–8 (2023).
8. Van Veen, D. *et al.* Adapted large language models can outperform medical experts in clinical text summarization. *Nat. Med.* **30**, 1134–1142 (2024).
9. Jiang, L. Y. *et al.* Health system-scale language models are all-purpose prediction engines. *Nature* **619**, 1–6 (2023).
10. Goh, E. *et al.* GPT-4 assistance for improvement of physician performance on patient care tasks: A randomized controlled trial. *Nat. Med.* 1–6 (2025) doi:10.1038/s41591-024-03456-y.
11. Jeblick, K. *et al.* ChatGPT makes medicine easy to swallow: an exploratory case study on simplified radiology reports. *Eur. Radiol.* <https://doi.org/10.1007/s00330-023-10213-1> (2023) doi:10.1007/s00330-023-10213-1.
12. Goodman, R. S. *et al.* Accuracy and Reliability of Chatbot Responses to Physician Questions. *JAMA Netw. Open* **6**, e2336483 (2023).
13. Sheng, B. *et al.* Large language models for diabetes care: Potentials and prospects. *Sci. Bull.* **69**, 583–588 (2024).
14. Zaretsky, J. *et al.* Generative artificial intelligence to transform inpatient discharge summaries to patient-friendly language and format. *JAMA Netw. Open* **7**, e240357 (2024).
15. Wang, X. *et al.* ChatGPT: promise and challenges for deployment in low- and middle-income countries. *Lancet Reg. Health – West. Pac.* **41**, (2023).
16. Ferryman, K., Mackintosh, M. & Ghassemi, M. Considering Biased Data as Informative Artifacts in AI-Assisted Health Care. *N. Engl. J. Med.* **389**, 833–838 (2023).
17. McCoy, L. G., Manrai, A. K. & Rodman, A. Large language models and the degradation of the medical record. *N. Engl. J. Med.* **391**, 1561–1564 (2024).
18. Raji, I. D., Daneshjou, R. & Alsentzer, E. It’s time to bench the medical exam benchmark. *NEJM AI* **2**, A1e2401235 (2025).
19. Jin, D. *et al.* What disease does this patient have? A large-scale open domain question answering dataset from medical exams. *Appl. Sci.* **11**, (2021).

20. Liévin, V., Hother, C. E., Motzfeldt, A. G. & Winther, O. Can large language models reason about medical questions? *Patterns* **5**, (2024).
21. Jin, Q., Dhingra, B., Liu, Z., Cohen, W. & Lu, X. PubMedQA: a dataset for biomedical research question answering. in *Proceedings of the 2019 conference on empirical methods in natural language processing and the 9th international joint conference on natural language processing (EMNLP-IJCNLP)* 2567–2577 (Association for Computational Linguistics, Hong Kong, China, 2019). doi:10.18653/v1/D19-1259.
22. Chen, Q. *et al.* Benchmarking large language models for biomedical natural language processing applications and recommendations. *Nat. Commun.* **16**, 3280 (2025).
23. Kreimeyer, K. *et al.* Natural language processing systems for capturing and standardizing unstructured clinical information: A systematic review. *J. Biomed. Inform.* **73**, 14–29 (2017).
24. OpenAI. GPT-4 Technical Report. Preprint at <https://doi.org/10.48550/arXiv.2303.08774> (2023).
25. Singhal, K. *et al.* Toward expert-level medical question answering with large language models. *Nat. Med.* 1–8 (2025).
26. Soroush, A. *et al.* Large language models are poor medical coders — benchmarking of medical code querying. *NEJM AI* **1**, AIdbp2300040 (2024).
27. Barile, J. *et al.* Diagnostic accuracy of a large language model in pediatric case studies. *JAMA Pediatr.* **178**, 313–315 (2024).
28. Arora, R. K. *et al.* HealthBench: Evaluating large language models towards improved human health. Preprint at <https://doi.org/10.48550/arXiv.2505.08775> (2025).
29. Goh, E. *et al.* Large language model influence on diagnostic reasoning: A randomized clinical trial. *JAMA Netw. Open* **7**, e2440969 (2024).
30. Nayak, A. *et al.* Comparison of History of Present Illness Summaries Generated by a Chatbot and Senior Internal Medicine Residents. *JAMA Intern. Med.* **183**, 1026–1027 (2023).
31. Sandmann, S. *et al.* Benchmark evaluation of DeepSeek large language models in clinical decision-making. *Nat. Med.* 1–1 (2025) doi:10.1038/s41591-025-03727-2.
32. Tordjman, M. *et al.* Comparative benchmarking of the DeepSeek large language model on medical tasks and clinical reasoning. *Nat. Med.* 1–6 (2025) doi:10.1038/s41591-025-03726-3.
33. Bedi, S. *et al.* MedHELM: Holistic evaluation of large language models for medical tasks. Preprint at <https://doi.org/10.48550/arXiv.2505.23802> (2025).
34. Qiu, P. *et al.* Towards building multilingual language model for medicine. *Nat. Commun.* **15**, 8384 (2024).
35. Zack, T. *et al.* Assessing the potential of GPT-4 to perpetuate racial and gender biases in health care: A model evaluation study. *Lancet Digit. Health* **6**, e12–e22 (2024).
36. Liu, X. *et al.* Uncovering Language Disparity of ChatGPT on Retinal Vascular Disease Classification: Cross-Sectional Study. *J. Med. Internet Res.* **26**, e51926 (2024).
37. Kim, J., Cai, Z. R., Chen, M. L., Simard, J. F. & Linos, E. Assessing Biases in Medical Decisions via Clinician and AI Chatbot Responses to Patient Vignettes. *JAMA Netw. Open* **6**, e2338050 (2023).
38. Zeng, Q. *et al.* GreenPLM: Cross-Lingual Transfer of Monolingual Pre-Trained Language Models at Almost No Cost. in *Proceedings of the Thirty-Second International Joint Conference on*

- Artificial Intelligence* 6290–6298 (International Joint Conferences on Artificial Intelligence Organization, Macau, SAR China, 2023). doi:10.24963/ijcai.2023/698.
39. Hofmann, V., Kalluri, P. R., Jurafsky, D. & King, S. AI generates covertly racist decisions about people based on their dialect. *Nature* **633**, 147–154 (2024).
 40. Ong, J. C. L. *et al.* Ethical and regulatory challenges of large language models in medicine. *Lancet Digit. Health* **6**, e428–e432 (2024).
 41. Ji, Y. *et al.* Mitigating the risk of health inequity exacerbated by large language models. *Npj Digit. Med.* **8**, 1–11 (2025).
 42. Chang, Y. *et al.* A Survey on Evaluation of Large Language Models. *ACM Trans. Intell. Syst. Technol.* <https://doi.org/10.1145/3641289> (2024) doi:10.1145/3641289.
 43. Singhal, K. *et al.* Large language models encode clinical knowledge. *Nature* **620**, 172–180 (2023).
 44. Xie, Q. *et al.* Medical foundation large language models for comprehensive text analysis and beyond. *Npj Digit. Med.* **8**, 1–10 (2025).
 45. Liu, X. *et al.* A generalist medical language model for disease diagnosis assistance. *Nat. Med.* 1–11 (2025) doi:10.1038/s41591-024-03416-6.
 46. Chiang, W.-L. *et al.* Chatbot arena: An open platform for evaluating LLMs by human preference. in *Forty-first international conference on machine learning* (2024).
 47. Minssen, T., Vayena, E. & Cohen, I. G. The Challenges for Regulating Medical Use of ChatGPT and Other Large Language Models. *JAMA* **330**, 315–316 (2023).
 48. Bedi, S. *et al.* Testing and Evaluation of Health Care Applications of Large Language Models: A Systematic Review. *JAMA* <https://doi.org/10.1001/jama.2024.21700> (2024) doi:10.1001/jama.2024.21700.
 49. Tang, Y.-D., Dong, E.-D. & Gao, W. LLMs in medicine: The need for advanced evaluation systems for disruptive technologies. *The Innovation* **5**, (2024).
 50. Wu, J. *et al.* Clinical Text Datasets for Medical Artificial Intelligence and Large Language Models — A Systematic Review. *NEJM AI* <https://doi.org/10.1056/AIra2400012> (2024) doi:10.1056/AIra2400012.
 51. DeepSeek-AI *et al.* DeepSeek-R1: Incentivizing reasoning capability in LLMs via reinforcement learning. Preprint at <https://doi.org/10.48550/arXiv.2501.12948> (2025).
 52. OpenAI *et al.* GPT-4o system card. Preprint at <https://doi.org/10.48550/arXiv.2410.21276> (2024).
 53. Gemini Team *et al.* Gemini 1.5: Unlocking multimodal understanding across millions of tokens of context. Preprint at <https://doi.org/10.48550/arXiv.2403.05530> (2024).
 54. Comanici, G. *et al.* Gemini 2.5: Pushing the Frontier with Advanced Reasoning, Multimodality, Long Context, and Next Generation Agentic Capabilities. Preprint at <https://doi.org/10.48550/arXiv.2507.06261> (2025).
 55. The llama 4 herd: The beginning of a new era of natively multimodal AI innovation. *Meta AI* <https://ai.meta.com/blog/llama-4-multimodal-intelligence/>.
 56. Yang, A. *et al.* Qwen3 Technical Report. Preprint at <https://doi.org/10.48550/arXiv.2505.09388> (2025).
 57. Brown, T. *et al.* Language Models are Few-Shot Learners. in *Advances in Neural Information Processing Systems* vol. 33 1877–1901 (Curran Associates, Inc., 2020).

58. Wei, J. *et al.* Chain-of-thought prompting elicits reasoning in large language models. in *Proceedings of the 36th International Conference on Neural Information Processing Systems* 24824–24837 (Curran Associates Inc., Red Hook, NY, USA, 2022).
59. Chen, J. *et al.* HuatuoGPT-o1, towards medical complex reasoning with LLMs. Preprint at <https://doi.org/10.48550/arXiv.2412.18925> (2024).
60. Sellergren, A. *et al.* MedGemma Technical Report. Preprint at <https://doi.org/10.48550/arXiv.2507.05201> (2025).
61. Team, B.-M. *et al.* Baichuan-M2: Scaling Medical Capability with Large Verifier System. Preprint at <https://doi.org/10.48550/arXiv.2509.02208> (2025).
62. Jiang, A. Q. *et al.* Mistral 7B. Preprint at <https://doi.org/10.48550/arXiv.2310.06825> (2023).
63. Qwen *et al.* Qwen2.5 technical report. Preprint at <https://doi.org/10.48550/arXiv.2412.15115> (2025).
64. Gemma Team *et al.* Gemma 3 technical report. Preprint at <https://doi.org/10.48550/arXiv.2503.19786> (2025).
65. Grattafiori, A. *et al.* The llama 3 herd of models. Preprint at <https://doi.org/10.48550/arXiv.2407.21783> (2024).
66. Nexusflow/athene-V2-chat · hugging face. <https://huggingface.co/Nexusflow/Athene-V2-Chat> (2024).
67. Wu, C. *et al.* Towards evaluating and building versatile large language models for medicine. *Npj Digit. Med.* **8**, 58 (2025).
68. OpenAI *et al.* gpt-oss-120b & gpt-oss-20b Model Card. Preprint at <https://doi.org/10.48550/arXiv.2508.10925> (2025).
69. Percha, B. Modern Clinical Text Mining: A Guide and Review. *Annu. Rev. Biomed. Data Sci.* **4**, 165–187 (2021).
70. Huang, K. *et al.* A foundation model for clinician-centered drug repurposing. *Nat. Med.* 1–13 (2024) doi:10.1038/s41591-024-03233-x.
71. Savage, T., Nayak, A., Gallo, R., Rangan, E. & Chen, J. H. Diagnostic reasoning prompts reveal the potential for large language model interpretability in medicine. *Npj Digit. Med.* **7**, 1–7 (2024).
72. Kasai, J., Kasai, Y., Sakaguchi, K., Yamada, Y. & Radev, D. Evaluating GPT-4 and ChatGPT on Japanese Medical Licensing Examinations. Preprint at <https://doi.org/10.48550/arXiv.2303.18027> (2023).
73. Pfohl, S. R. *et al.* A toolbox for surfacing health equity harms and biases in large language models. *Nat. Med.* **30**, 3590–3600 (2024).
74. DeepSeek-AI *et al.* DeepSeek LLM: Scaling open-source language models with longtermism. Preprint at <https://doi.org/10.48550/arXiv.2401.02954> (2024).
75. Buckley, T. A., Crowe, B., Abdunour, R.-E. E., Rodman, A. & Manrai, A. K. Comparison of frontier open-source and proprietary large language models for complex diagnoses. *JAMA Health Forum* **6**, e250040 (2025).
76. Yang, X. *et al.* A large language model for electronic health records. *Npj Digit. Med.* **5**, 1–9 (2022).
77. Hager, P. *et al.* Evaluation and mitigation of the limitations of large language models in clinical decision-making. *Nat. Med.* **30**, 2613–2622 (2024).

78. Wornow, M. *et al.* The shaky foundations of large language models and foundation models for electronic health records. *Npj Digit. Med.* **6**, 1–10 (2023).
79. Meshkin, H. *et al.* Harnessing large language models' zero-shot and few-shot learning capabilities for regulatory research. *Brief. Bioinform.* **25**, bbae354 (2024).
80. Agrawal, M., Hegselmann, S., Lang, H., Kim, Y. & Sontag, D. Large language models are few-shot clinical information extractors. in *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing 1998–2022* (Association for Computational Linguistics, Abu Dhabi, United Arab Emirates, 2022).
81. Qin, L. *et al.* A survey of multilingual large language models. *Patterns* **6**, (2025).
82. Liu, F. *et al.* A multimodal multidomain multilingual medical foundation model for zero shot clinical diagnosis. *Npj Digit. Med.* **8**, 1–12 (2025).
83. Das, B. C., Amini, M. H. & Wu, Y. Security and privacy challenges of large language models: A survey. *ACM Comput Surv* **57**, 152:1-152:39 (2025).
84. Lehman, E. *et al.* Do we still need clinical language models? in *Proceedings of the Conference on Health, Inference, and Learning* (2023).
85. Wang, X. *et al.* Self-consistency improves chain of thought reasoning in language models. Preprint at <https://doi.org/10.48550/arXiv.2203.11171> (2023).
86. Wu, J. *et al.* Large language models leverage external knowledge to extend clinical insight beyond language boundaries. *J. Am. Med. Inform. Assoc.* ocae079 (2024) doi:10.1093/jamia/ocae079.
87. Wu, J., Wu, X. & Yang, J. Guiding clinical reasoning with large language models via knowledge seeds. in vol. 8 7491–7499 (2024).
88. Qwen Team. QwQ-32B: Embracing the power of reinforcement learning. *Qwen* <https://qwenlm.github.io/blog/qwq-32b/> (2025).
89. Zakka, C. *et al.* Almanac — Retrieval-Augmented Language Models for Clinical Medicine. *NEJM AI* **1**, AIoa2300068 (2024).
90. Shah, N. H., Entwistle, D. & Pfeffer, M. A. Creation and Adoption of Large Language Models in Medicine. *JAMA* <https://doi.org/10.1001/jama.2023.14217> (2023) doi:10.1001/jama.2023.14217.
91. Karargyris, A. *et al.* Federated benchmarking of medical artificial intelligence with MedPerf. *Nat. Mach. Intell.* **5**, 799–810 (2023).
92. Rodman, A., Zwaan, L., Olson, A. & Manrai, A. K. When it comes to benchmarks, humans are the only way. *NEJM AI* **2**, AIe2500143 (2025).
93. Gallifant, J. *et al.* The TRIPOD-LLM reporting guideline for studies using large language models. *Nat. Med.* **31**, 60–69 (2025).
94. Gu, B., Desai, R. J., Lin, K. J. & Yang, J. Probabilistic medical predictions of large language models. *Npj Digit. Med.* **7**, 1–9 (2024).
95. National NLP clinical challenges (n2c2). <https://n2c2.dbmi.hms.harvard.edu/home>.
96. CLEF eHealth lab series. <https://clefehealth.imag.fr/clefehealth.imag.fr/index.html>.
97. Goldberger, A. L. *et al.* PhysioBank, PhysioToolkit, and PhysioNet: Components of a New Research Resource for Complex Physiologic Signals¹. *Circulation* **101**, e215–e220 (2000).
98. Hugging face – the AI community building the future. <https://huggingface.co/> (2025).
99. Kwon, W. *et al.* Efficient memory management for large language model serving with PagedAttention. in *Proceedings of the 29th Symposium on Operating Systems Principles* 611–626

- (Association for Computing Machinery, New York, NY, USA, 2023). doi:10.1145/3600006.3613165.
100. Lin, C.-Y. ROUGE: A Package for Automatic Evaluation of Summaries. in *Text Summarization Branches Out* 74–81 (Association for Computational Linguistics, Barcelona, Spain, 2004).
 101. Papineni, K., Roukos, S., Ward, T. & Zhu, W.-J. BLEU: A method for automatic evaluation of machine translation. in *Proceedings of the 40th Annual Meeting on Association for Computational Linguistics - ACL '02* 311 (Association for Computational Linguistics, Philadelphia, Pennsylvania, 2001). doi:10.3115/1073083.1073135.
 102. Zhang, T., Kishore, V., Wu, F., Weinberger, K. Q. & Artzi, Y. BERTScore: Evaluating text generation with BERT. in *International conference on learning representations* (2020).
 103. Pedregosa, F. *et al.* Scikit-learn: Machine learning in python. *J. Mach. Learn. Res.* **12**, 2825–2830 (2011).
 104. Bird, S. NLTK: The natural language toolkit. in *Proceedings of the COLING/ACL 2006 Interactive Presentation Sessions* 69–72 (Association for Computational Linguistics, Sydney, Australia, 2006). doi:10.3115/1225403.1225421.
 105. Google Research. Python ROUGE Implementation. *GitHub* <https://github.com/google-research/google-research/tree/master/rouge>.
 106. Zhang, T., Kishore, V., Wu, F., Weinberger, K. Q. & Artzi, Y. BERTScore: Evaluating text generation with BERT. Preprint at <https://doi.org/10.48550/arXiv.1904.09675> (2020).
 107. Xu, R., Wang, Z., Fan, R.-Z. & Liu, P. Benchmarking benchmark leakage in large language models. Preprint at <https://doi.org/10.48550/arXiv.2404.18824> (2024).